Md Kamrul Islam

Lecturer in Computer Science | AI Research Engineer

Dhaka, Bangladesh | +8801864208599 | mdkamrul.islam@aiub.edu

Google Scholar | ORCID | Portfolio | GitHub | LinkedIn

Research Profile

AI researcher with a background in deep learning and medical imaging, currently focused on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and knowledge-grounded AI. My emerging research direction explores Multimodal Large Language Models (MLLMs) for building reliable and knowledge-grounded AI systems.

Education

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA) — *Sept 2023 – Oct 2025*
CentraleSupélec | Universitat Politècnica de Catalunya (UPC) | Université Libre de Bruxelles (ULB) — *France | Spain | Belgium*

- **Final Grade:** 14.5/20 (*Bien*)
- **Thesis:** *A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models*

Bachelor of Engineering in Software Engineering — *Mar 2018 – Dec 2021*
Sichuan University — *Chengdu, China*

- **CGPA:** 3.64/4.00 (Weighted Average: 87%)
- **Thesis:** *Brain Tumor Detection and Classification Using Convolutional Neural Networks*

Research Experience

AI Research Engineer Intern [Project] — *May 2025 – Oct 2025*
LISSI, Université Paris-Est Créteil | in collaboration with CEA–LIST — *Paris, France*

- Developed a hybrid framework combining LLM-based reasoning with rule-based validation for automated security annotation generation in Business Process Model and Notation (BPMN).
- Designed a benchmark dataset and evaluated multiple LLMs using different prompt settings and RAG for security annotation generation.
- Achieved higher annotation accuracy while reducing annotation time by 95% compared with domain experts.

Methods & Tools: *LLMs, RAG, Python, LangChain, Flask, React.js, CI/CD*

Graduate Research Assistant [Project] — *Oct 2024 – Feb 2025*
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec — *Paris, France*

- Developed an unsupervised deep-learning framework combining Convolutional Neural Networks (CNNs) and Group-Equivariant CNNs (G-CNNs) for representation learning and clustering of large-scale medical imaging data.
- Investigated geometric inductive biases to improve representation robustness and reduce reliance on data augmentation.
- Implemented multi-GPU distributed training pipelines using PyTorch, CUDA, and HPC infrastructure for scalable experimentation.

Methods & Tools: *PyTorch, G-CNNs, OpenCV, CUDA, HPC*

Publications

1. **M. K. Islam**, T. Henry, M. Salnitri, J. Köpke, and S. Souihi, "*A Hybrid LLM-Based Framework for Automated Security Annotations Generation in Business Process Models*," *24th International Conference on Business Process Management (BPM 2026) – BPM Forum*, 2026. **Accepted.** [Preprint]
2. C. K. Sah, L. Xiaoli, M. M. Islam, and **M. K. Islam**, "*Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education*," *2024 36th International Conference on Software Engineering Education and Training (CSEE&T)*, Würzburg, Germany, 2024, pp. 1–5. [DOI]

Academic & Professional Experience

Lecturer [Faculty Profile] *April 2026 – Present*
Department of Computer Science, American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- Teach undergraduate courses in Data Visualization, Data Structures, and Programming for Data Science.
- Contribute to curriculum and course development aligned with international standards.
- Supervise undergraduate research projects in problem formulation, experimentation, and scientific writing.

Big Data Engineer Intern *Dec 2020 – May 2021*
Chengdu Suncape Co., Ltd. *Chengdu, China*

- Developed an end-to-end data pipeline for large-scale data ingestion, transformation, and analytics in distributed environments.
- Enhanced data preprocessing and feature engineering, improving predictive model accuracy by 20%.
- Collaborated in an Agile Scrum environment to implement modular software design and CI/CD practices.

Methods & Tools: *Apache Spark, Scikit-learn, SQL, Git, Jira, CI/CD*

Technical Skills

Programming: Python, C/C++, SQL, JavaScript
ML/AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, LLM APIs, LangChain
Big Data & Cloud: Apache Spark, Microsoft Azure, Microsoft Fabric, Apache Airflow
Databases: PostgreSQL, Neo4j, GraphDB, SPARQL
Computing: Linux, Docker, CUDA, HPC, Git

Projects

DigiScan360 [Project] *Feb 2024 – June 2024*
Developed a competitive intelligence platform and pitched it as a startup prototype.
Tech Stack: *PySpark, LLMs, Microsoft Fabric, Microsoft Azure, Power BI, GraphDB, SPARQL*

Anomaly Detection in Diesel Train Cooling Systems [Project] *Sept 2023 – Dec 2023*
Developed unsupervised models to detect anomalies from spatio-temporal data in train cooling systems.
Tech Stack: *Python, SQL, Scikit-learn, Tableau*

Academic Activities & Service

Reviewer – 4th International Conference on Computing Advancements (ICCA 2026) [Certificate] *Aug 2026*
ACM SIGAPP In-Cooperation | American International University-Bangladesh *Dhaka, Bangladesh*

Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) [Poster] *July 2024*
University of Padova *Padova, Italy*

Honors & Scholarships

Erasmus Mundus Joint Master Degree Scholarship *2023 – 2025*
European Commission – Fully funded merit scholarship for the BDMA Joint Master's Programme

Belt and Road Initiative Scholarship *2018 – 2022*
Chinese Government – Fully funded scholarship for undergraduate study at Sichuan University

Languages

English: C2 (Proficient) | **Chinese:** B2 (Upper-intermediate) | **French:** A2 (Elementary) | **Bengali:** Native

References

References available upon request.